

Task 1 - Problem Definition

What is the real-world problem? - Climate change is a dangerous and ever-changing phenomenon regarding our world's temperature. Patterns in emissions, sea levels, rainfall, and temperature are affected by climate change. Despite decades of data, there are unclear patterns in how different countries contribute and in the factors driving changes in the climate; we plan to use data to help solve these patterns.

Why does it matter? - Rising temperatures and emissions threaten agricultural operations, the ecosystem in both land and sea, and overall public health. Being able to predict climate change trends can help mitigate the damage caused by these issues. Farmers can plan for sudden changes in the climate, and specialists can help preserve certain ecosystems if given enough warning. Being able to resolve this issue before irreversible damage occurs is important.

Who are the stakeholders? - Some of the stakeholders would include Climate scientists and researchers, agricultural communities and farmers, the government, policymakers for limiting actions that cause the most climate change, Coastal wildlife managers, and fish farmers.

What Decisions will be made from the results? - From the data, we can pinpoint certain high-impact regions to prioritize the reduction of the high-impact changes. New rules and regulations can be put in place to combat climate change, driven by the factors that contribute most. Stricter regulations can be implemented in the countries with the highest impact. We can help predict how the change will affect future temperatures and rainfall.